

Linear Algebra Done Right

Reading Notes

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1 Vector Spaces

1.1 $\mathbb{R}^n$ and $\mathbb{C}^n$

Before defining vector spaces, Axler establishes the fields we are working over. The symbol $\mathbb{F}$ will always denote either the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$.

Definition 1.1 (Fields). *A field is a set containing at least two distinct elements called 0 and 1, along with operations of addition and multiplication satisfying the properties of commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity.*

1.1.1 Complex Numbers, $\mathbb{C}$

Definition 1.2 (Complex Numbers).

- A complex number is an ordered pair (a, b) where $a, b \in \mathbb{R}$, but we will write this as $a + bi$.
- The set of all complex numbers is denoted by $\mathbb{C}$:

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

- Addition and multiplication on $\mathbb{C}$ are defined by

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i;$$

here $a, b, c, d \in \mathbb{R}$

My thoughts

Just assume $i^2 = -1$ and everything works the same as in $\mathbb{R}$. Commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity are all properties of complex arithmetic.

1.1.2 Lists

Definition 1.3 (List, length).

- Suppose n is a nonnegative integer. A list of length n is an ordered collection of n elements (which might be numbers, other lists, or more abstract objects).
- Two lists are equal if and only if they have the same length and the same elements in the same order.

My thoughts

A list can have multiple same entries unlike sets. Fields and Vector Spaces (defined later) are sets, but a list is a tuple.

1.1.3 $\mathbb{F}^n$

Definition 1.4 ($\mathbb{F}^n$, coordinate). $\mathbb{F}^n$ is the set of all lists of length n of elements of $\mathbb{F}$:

$$\mathbb{F}^n = \{(x_1, \dots, x_n) : x_k \in \mathbb{F} \text{ for } k \in \{1, \dots, n\}\}$$

For $(x_1, \dots, x_n) \in \mathbb{F}^n$ and $k \in \{1, \dots, n\}$, we say that x_k is the k^{th} coordinate of $(x_1, \dots, x_n)$.

Definition 1.5 (addition in F^n). *Addition in F^n is defined by adding corresponding coordinates:*

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

Definition 1.6 (additive inverse in $\mathbb{F}^n$, $-x$). *For $x \in \mathbb{F}^n$, the additive inverse of x , denoted by $-x$, is the vector $-x$ in $\mathbb{F}^n$ such that*

$$x + (-x) = 0$$

Thus if $x = (x_1, \dots, x_n)$, then $-x = (-x_1, \dots, -x_n)$.

Definition 1.7 (scalar multiplication in $\mathbb{F}^n$). *The product of a number λ and a vector in $\mathbb{F}^n$ is computed by multiplying each coordinate of the vector by λ :*

$$\lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n);$$

here $\lambda \in \mathbb{F}$ and $(x_1, \dots, x_n) \in \mathbb{F}^n$

1.2 Definition of a Vector Space

Now we abstract the properties of $\mathbb{R}^n$ and $\mathbb{C}^n$ into a general structure.

Definition 1.8 (Vector Space). *A vector space is a set $\mathcal{V}$ along with an addition on $\mathcal{V}$ and a scalar multiplication on $\mathcal{V}$ such that commutativity, associativity, identities, additive inverse, and distributivity hold.*

My thoughts

Essentially, the difference between a field and a vector space is the fact that only scalar multiplication is defined for elements of a vector space. This also implies it is not necessary for a vector space to have multiplicative inverses. Also note, a field is a standalone set while a vector space relies on an underlying field to exist at all.

Definition 1.9 (Real Vector Space, Complex Vector Space).

- *A vector space over $\mathbb{R}$ is called a real vector space.*
- *A vector space over $\mathbb{C}$ is called a complex vector space.*

1.2.1 Some Property Proofs

Theorem 1.10 (Unique Additive Identity). *A vector space has a unique additive identity.*

Proof. Suppose 0 and $0'$ are both additive identities for V . Then

$$0' = 0' + 0 = 0 + 0' = 0,$$

where the first equality holds because 0 is an additive identity, the second equality comes from commutativity, and the third equality holds because $0'$ is an additive identity. Thus $0' = 0$, proving that $\mathcal{V}$ has only one additive identity. $\square$

Theorem 1.11. *a number times the vector 0 is 0*

Proof. For $v \in \mathcal{V}$, we have $0v = (0 + 0)v = 0v + 0v$. Adding the additive inverse of $0v$ to both sides of the equation above gives $0 = 0v$, as desired. $\square$

Theorem 1.12. *the number -1 times a vector gives the additive inverse of the vector*

Proof. For $v \in \mathcal{V}$, we have $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$. This equation says that $(-1)v$, when added to v , gives 0 . Thus $(-1)v$ is the additive inverse of v , as desired $\square$

1.3 Subspaces

Definition 1.13 (Subspace). A subset U of $\mathcal{V}$ is called a subspace of $\mathcal{V}$ if U is also a vector space under the same addition and scalar multiplication.

1.3.1 Sums of Subspaces

Definition 1.14 (Sum of Subspaces). Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$. The sum of $\mathcal{V}_1, \dots, \mathcal{V}_m$, denoted by $\mathcal{V}_1 + \dots + \mathcal{V}_m$, is the set of all possible sums of elements of $\mathcal{V}_1, \dots, \mathcal{V}_m$. More precisely,

$$\mathcal{V}_1 + \dots + \mathcal{V}_m = \{v_1 + \dots + v_m : v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m\}$$

Theorem 1.15. Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$.

My thoughts

I will add my own proof that sum of subspaces is a subspace (although Axler skips this) because I think it is necessary.

Proof. Let us first prove that sum of subspaces is a subspace

- $\mathcal{V}_1 + \dots + \mathcal{V}_m$ contains 0 (additive identity) since all the subspaces contain 0 and sum of subspaces contain every possible sum of their elements.
- Let $x, y \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$.
By definition we can express them as $x = v_1 + \dots + v_m$ and $y = u_1 + \dots + u_m$ such that $u_1, v_1 \in \mathcal{V}_1, \dots, u_m, v_m \in \mathcal{V}_m$. So we have

$$x + y = (v_1 + \dots + v_m) + (u_1 + \dots + u_m)$$

By commutativity and associativity of vector addition,

$$x + y = (v_1 + u_1) + \dots + (v_m + u_m)$$

Since $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces, $(v_1 + u_1) \in \mathcal{V}_1, \dots, (v_m + u_m) \in \mathcal{V}_m$

Hence we conclude $(x + y) \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, it is closed under addition

- Let $x \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$. and $x = v_1 + \dots + v_m$ where $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$. Let c be any scalar from the field $\mathbb{F}$.
Multiplying x with the scalar c yields:

$$cx = c(v_1 + \dots + v_m) = cv_1 + \dots + cv_m$$

Since $\mathcal{V}_1 + \dots + \mathcal{V}_m$ are subspaces $cv_1 \in \mathcal{V}_1, \dots, cv_m \in \mathcal{V}_m$.

We conclude $cx \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, sum of subspaces is a subspace.

Now we can start with the proof for $\mathcal{V}_1, \dots, \mathcal{V}_m$ being the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$.

The subspaces $\mathcal{V}_1, \dots, \mathcal{V}_m$ are all contained in $\mathcal{V}_1 + \dots + \mathcal{V}_m$, considering sums $v_1 + \dots + v_m$ where all except one of the v_k 's are 0.

Conversely, every subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$ contains $\mathcal{V}_1 + \dots + \mathcal{V}_m$ (because subspaces must contain all finite sums of their elements). Thus $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$. $\square$

1.3.2 Direct Sums

Definition 1.16 (Direct Sum). Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$.

- The sum $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is called a *direct sum* if each element of $\mathcal{V}_1 + \dots + \mathcal{V}_m$ can be written in only one way as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$.
- If $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum, then $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_m$ denotes $\mathcal{V}_1 + \dots + \mathcal{V}_m$, with the $\oplus$ notation serving as an indication that this is a direct sum.

Theorem 1.17. Suppose $\mathcal{V}_1 + \dots + \mathcal{V}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum if and only if the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0.

Proof. First suppose $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum. Then the definition of direct sum implies that the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0.

Now suppose that the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0. To show that $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum, let $v \in \mathcal{V}_1 + \dots + \mathcal{V}_m$. We can write

$$v = v_1 + \dots + v_m$$

for some $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$. To show that this representation is unique, suppose we also have

$$v = u_1 + \dots + u_m$$

where $u_1 \in \mathcal{V}_1, \dots, u_m \in \mathcal{V}_m$. Subtracting these two equations, we have

$$0 = (v_1 - u_1) + \dots + (v_m - u_m)$$

Because $v_1 - u_1 \in \mathcal{V}_1, \dots, v_m - u_m \in \mathcal{V}_m$, the equation above implies that each $v_k - u_k$ equals 0. Thus $v_1 = u_1, \dots, v_m = u_m$, as desired. $\square$

Theorem 1.18. Suppose U and W are subspaces of V . Then

$$U + W \text{ is a direct sum} \iff U \cap W = \{0\}.$$

Proof. $\square$